

1. **Create Operation:**

- Create a new database named "library" and switch to it.
- Create a collection named "books" with fields: title, author, and published_year.
- Insert a document into the "books" collection with the details of your favorite book.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\davee>mongosh
Current Mongosh Log ID: 69f786a4dbef22e4a3682d0
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
Using MongoDB:      8.2.7
Using Mongosh:       2.8.2
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-05-02T21:52:24.969+02:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

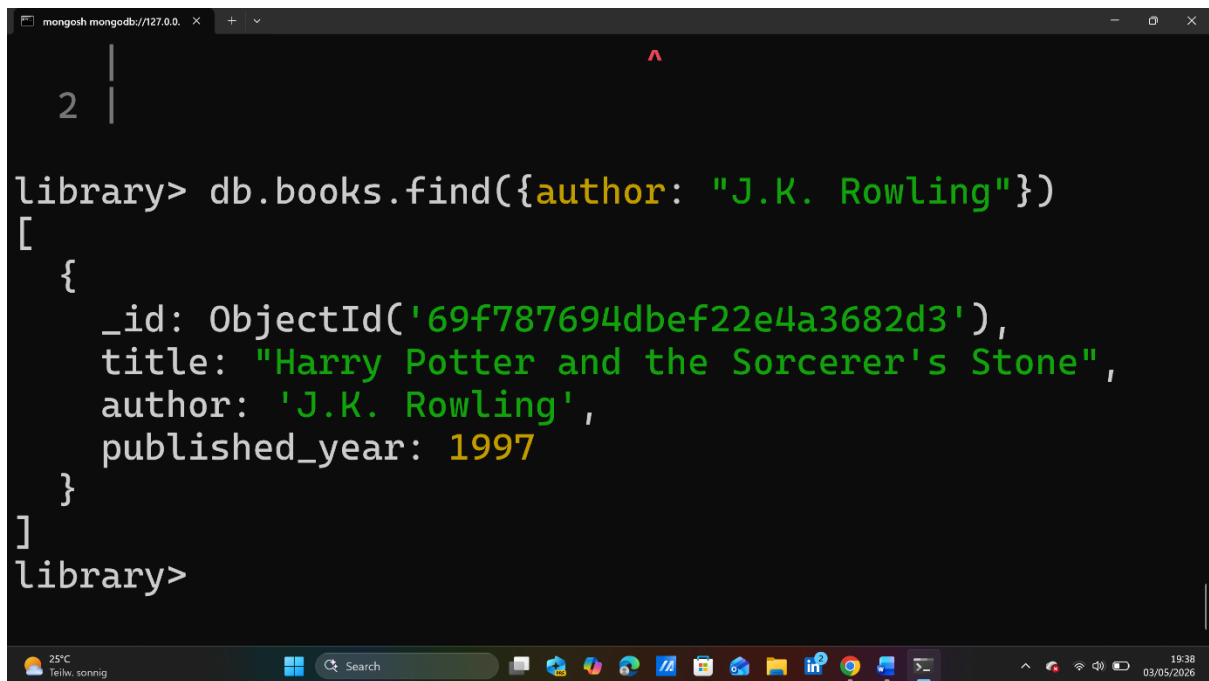
test> use library
switched to db library
library> db.createCollection("books")
{ ok: 1 }
library> db.books.insertOne({
  title: "The Great Gatsby",
  author: "F. Scott Fitzgerald",
  Published_year: 1925
})
{
  acknowledged: true,
  insertedId: ObjectId('69f786a84dbef22e4a3682d1')
}
library>
```

2. **Read Operation:**

- Retrieve all documents from the "books" collection.

```
}
library> db.books.find()
[
  {
    _id: ObjectId('69f786a84dbef22e4a3682d1'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    Published_year: 1925
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    published_year: 1925
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d3'),
    title: 'Harry Potter and the Sorcerer's Stone',
    author: 'J.K. Rowling',
    published_year: 1997
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d4'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 1951
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d5'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    published_year: 1960
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d6'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d7'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d8'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021
  }
]
```

- Find and display only the documents where the author is "J.K. Rowling".

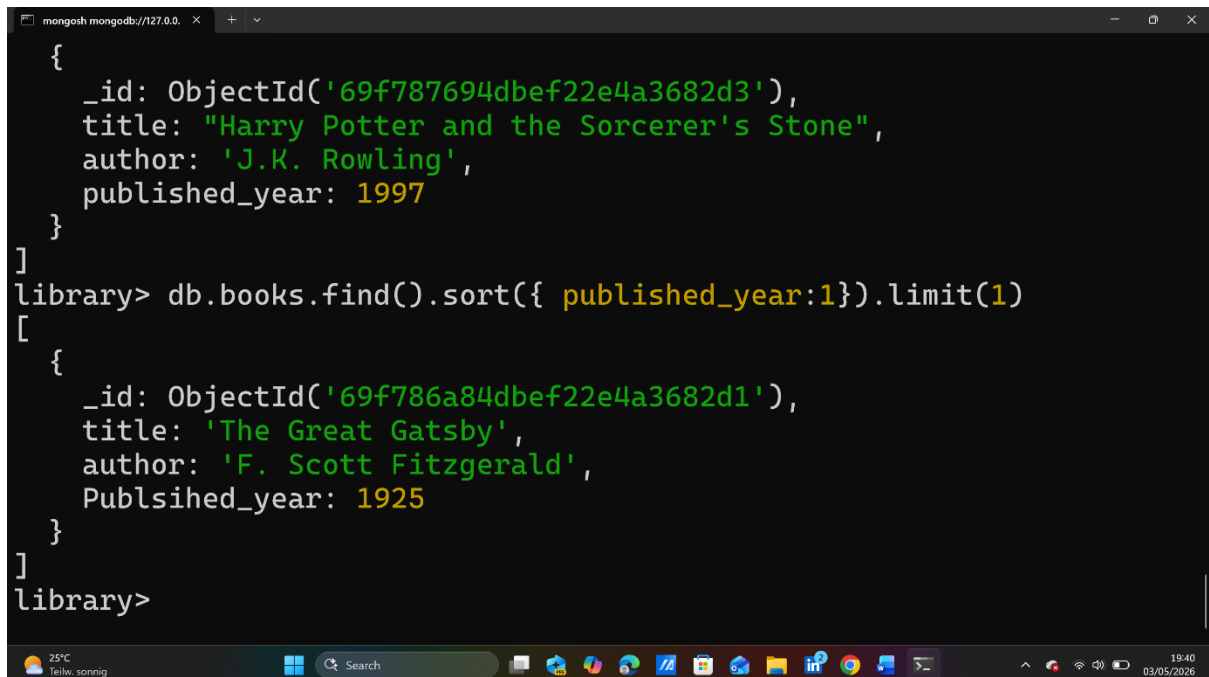


The screenshot shows a MongoDB terminal window with the following content:

```
library> db.books.find({author: "J.K. Rowling"})
[
  {
    _id: ObjectId('69f787694dbef22e4a3682d3'),
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    published_year: 1997
  }
]
library>
```

The terminal window has a title bar that reads "mongoosh mongodbat://127.0.0.1". The Windows taskbar at the bottom shows the date and time as 19:38 on 03/05/2026.

-Fetch and display the document with the earliest published year.



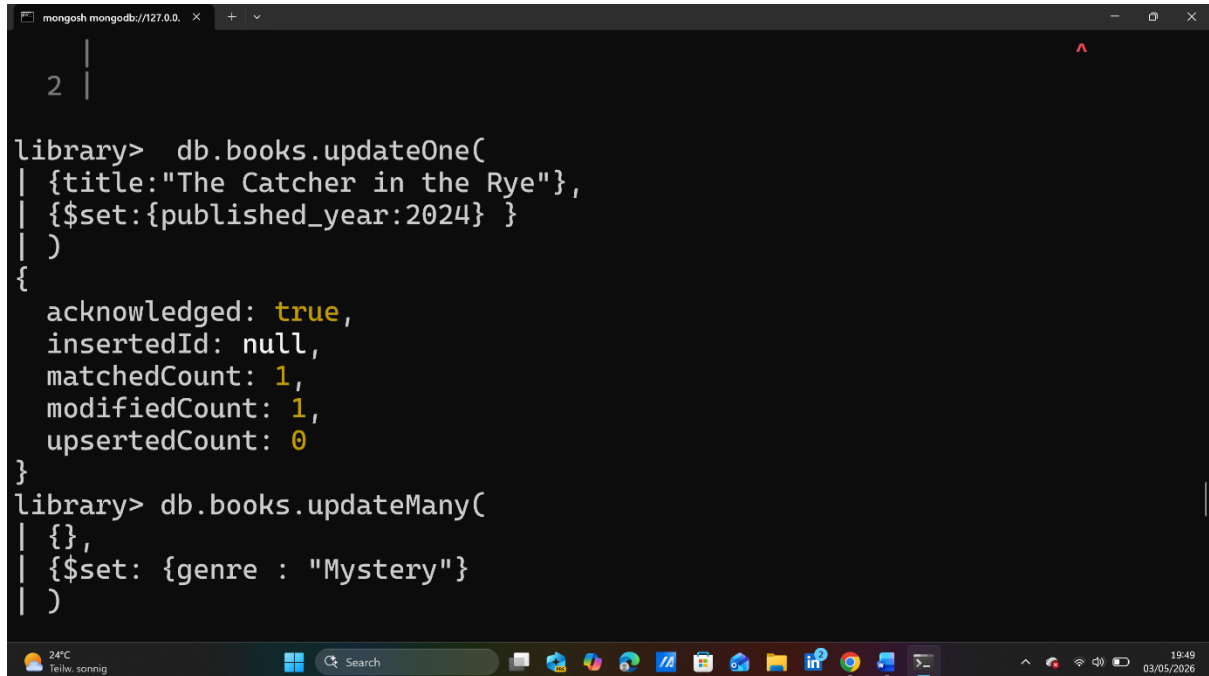
The screenshot shows a MongoDB terminal window with the following content:

```
{
  _id: ObjectId('69f787694dbef22e4a3682d3'),
  title: "Harry Potter and the Sorcerer's Stone",
  author: 'J.K. Rowling',
  published_year: 1997
}
]
library> db.books.find().sort({ published_year:1}).limit(1)
[
  {
    _id: ObjectId('69f786a84dbef22e4a3682d1'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    Published_year: 1925
  }
]
library>
```

The terminal window has a title bar that reads "mongoosh mongodbat://127.0.0.1". The Windows taskbar at the bottom shows the date and time as 19:40 on 03/05/2026.

3. **Update Operation:**

- Update the published year of the book with the title "The Catcher in the Rye" to the current year.
- Add a new field "genre" with the value "Mystery" to all documents in the "books" collection.

A screenshot of a MongoDB terminal window. The window title is 'mongosh mongodbs://127.0.0.1:27017'. The terminal shows two commands and their results. The first command is 'db.books.updateOne({title: "The Catcher in the Rye"}, {\$set: {published_year: 2024}})', which returns a JSON object with 'acknowledged: true', 'insertedId: null', 'matchedCount: 1', 'modifiedCount: 1', and 'upsertedCount: 0'. The second command is 'db.books.updateMany({}, {\$set: {genre: "Mystery"}})', which is partially visible. The terminal is running on a Windows desktop, with the taskbar at the bottom showing various application icons and the system clock indicating 15:48 on 03/05/2026.

```
mongosh mongodbs://127.0.0.1:27017
2 |
library> db.books.updateOne(
| {title:"The Catcher in the Rye"},
| {$set:{published_year:2024}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> db.books.updateMany(
| {},
| {$set: {genre : "Mystery"}}
| )
```

```

library> db.books.updateMany(
|   {},
|   { $set: { genre: "Mystery" } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 8,
  upsertedCount: 0
}
library> db.books.find()
[
  {
    _id: ObjectId('69f786a84dbef22e4a3682d1'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    Published_year: 1925,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d2'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    published_year: 1925,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d3'),
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    published_year: 1997,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d4'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2024,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d5'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    published_year: 1960,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d6'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d7'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d8'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library>

```

4. **Delete Operation:**

- Remove the document with the title "1984" from the "books" collection.
- Delete all documents from the "books" collection where the published year is before 2000.

```
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d8'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library> db.books.deleteOne({title: "1984"})
{ acknowledged: true, deletedCount: 1 }
library> db.books.deleteMany({ published_year: {$lt : 2000}})
{ acknowledged: true, deletedCount: 3 }
library>
```

5. **Advanced Query:**

- Find and display the top 3 recently published books from the "books" collection
- Retrieve documents from the "books" collection where the title contains the word "MongoDB" or "NoSQL".

```
mongosh mongodb://127.0.0.1:27020
> use library
library> db.books.find({
  published_year: 2021,
  genre: 'Mystery'
})
[
  {
    _id: ObjectId('69f787694dbef22e4a3682d7'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  }
]
library>
```

```
mongosh mongodb://127.0.0.1:27020
> use library
library> db.books.deleteOne({title: "1984"})
{ acknowledged: true, deletedCount: 1 }
library> db.books.deleteMany({ published_year: {$lt : 2000}})
{ acknowledged: true, deletedCount: 3 }
library> db.books.find().sort({published_year:-1}).limit(3)
[
  {
    _id: ObjectId('69f787694dbef22e4a3682d4'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2024,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d7'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f787694dbef22e4a3682d8'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library>
```